## Type MS · Classify immunity as natural or artificial and active or passive

### Answer 1 of 7 · Question 4 part a.i · 2 marks

2022 JAN · Paper 02 · extract 1 of 2 · find part a.i in this mark scheme extract

| Number | | | | |
|---|---|---|---|---|
| 4 | (a) (i) | method 1 = passive;<br>method 2 = active; | | 2 |
| | (ii) | • method 1 because antibodies injected;<br>• immediate high level in blood/faster response;<br>• to destroy, pathogen/disease organism; | Do not allow destroy / kill/attack disease | 3 |
| | (b) | • catch disease/pathogens enter body;<br>• antibodies produced;<br>• by white blood cells/lymphocytes;<br>• memory cells produced;<br>• rapid production of antibodies on re-infection/faster secondary immune response; | | 5 |

Total 10 marks

## Type MS · Classify immunity as natural or artificial and active or passive

### Answer 1 of 7 · Question 4 part a.i · 2 marks

2022 JAN · Paper 02 · extract 2 of 2 · find part a.i in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

## Type MS · Classify immunity as natural or artificial and active or passive

### Answer 2 of 7 · Question 2 part a.v · 2 marks

2020 NOV · Paper 02 (regional R) · extract 1 of 2 · find part a.v in this mark scheme extract

| Number | | | | | |
|---|---|---|---|---|---|
| 2 | (a) | (i) | • genetic material made of RNA (1)<br>• no DNA (1) | | 2 |
| | | (ii) | <table><tr><th>Disease</th><th>Blood Tested (✓)</th></tr><tr><td>anaemia</td><td></td></tr><tr><td>cystic fibrosis</td><td></td></tr><tr><td>gonorrhoea</td><td>✓</td></tr><tr><td>HIV</td><td>✓</td></tr><tr><td>scurvy</td><td></td></tr></table> | -1 for each extra tick | 2 |
| | | (iii) | • donor/blood transfused into person with Ebola (1)<br>• needs to be compatible/matched/same group/not rejected (1)<br>• otherwise agglutination/clumping occurs (1) | R clotting | 3 |
| | | (iv) | • cause blood cells to burst/cells are damaged/destroyed/lose structure (1)<br>• results in loss of function (1) | | 2 |
| | | (v) | • antibodies in donated blood (1)<br>• can destroy virus in infected person (1) | | 2 |
| | (b) | | • uracil present, not thymine (1)<br>• guanine would pair with cytosine/adenine with uracil (1)<br>• percentage of G and C/A and U would be the same (1) | | 3 |

Total for question = 14 marks

## Type MS · Classify immunity as natural or artificial and active or passive

### Answer 2 of 7 · Question 2 part a.v · 2 marks

2020 NOV · Paper 02 (regional R) · extract 2 of 2 · find part a.v in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

Type MS · Classify immunity as natural or artificial and active or passive

Answer 3 of 7 · Question 7 part b · 6 marks

2020 NOV · Paper 02 · extract 1 of 2 · find part b in this mark scheme extract

| | | | |
|---|---|---|---|
| 7 (a) | • plasma contains no cells (1)<br>• plasma colourless/whole blood red (1) | Allow reverse for whole blood | 2 |
| (b) | any six from<br>• blood of patient contains antibodies (1)<br>• (antibodies) bind to Ebola/virus antigens/antibody-antigen complex formed/antibodies complementary to (virus) antigens (1)<br>• phagocytes engulf (antibody-antigen complex) (1)<br>• Ebola/virus destroyed (1)<br>• testing of patient's blood ensures no disease present/prevents transfer of disease (1)<br>• removal of red blood cells ensures no reaction when blood transfused/no agglutination (1)<br>• anti-Ebola antibodies in transfused blood help person with disease to recover (1)<br>• more effective than medicine (1)<br>• quicker effect (1) | Ignore fight disease/virus<br><br>Allow passive artificial immunity | 6 |

Total 8 marks

## Type MS · Classify immunity as natural or artificial and active or passive

### Answer 3 of 7 · Question 7 part b · 6 marks

2020 NOV · Paper 02 · extract 2 of 2 · find part b in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|
| | | | |

## Type MS · Classify immunity as natural or artificial and active or passive

### Answer 4 of 7 · Question 4 part a.ii · 3 marks

2022 JAN · Paper 02 · extract 1 of 2 · find part a.ii in this mark scheme extract

| Number | | | | |
|---|---|---|---|---|
| 4 | (a) (i) | method 1 = passive;<br>method 2 = active; | | 2 |
| | (ii) | • method 1 because antibodies injected;<br>• immediate high level in blood/faster response;<br>• to destroy, pathogen/disease organism; | Do not allow destroy / kill/attack disease | 3 |
| | (b) | • catch disease/pathogens enter body;<br>• antibodies produced;<br>• by white blood cells/lymphocytes;<br>• memory cells produced;<br>• rapid production of antibodies on re-infection/faster secondary immune response; | | 5 |

Total 10 marks

## Type MS · Classify immunity as natural or artificial and active or passive

### Answer 4 of 7 · Question 4 part a.ii · 3 marks

2022 JAN · Paper 02 · extract 2 of 2 · find part a.ii in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

Type MS · Classify immunity as natural or artificial and active or passive

Answer 5 of 7 · Question 3 part b.ii · 2 marks

2023 NOV · Paper 02 · extract 1 of 2 · find part b.ii in this mark scheme extract

| Number | | | | |
|---|---|---|---|---|
| 3 | (a) | pathogens; | | 1 |
| | (b) (i) | 1.2 years; | | 1 |
| | (ii) | • active;<br>• artificial; | | 2 |
| | (iii) | Any two from<br>• weakened/dead/attenuated organisms;<br>• toxins;<br>• antigens;<br>• RNA; | Allow named bacterium/virus<br>Ignore genetic material | 2 |
| | (c) | Any two from<br>• passed from mother;<br>• during breastfeeding/from breast milk;<br>• across placenta; | Ignore umbilical cord | 2 |
| | (d) | Any five from<br>• antibody level falls;<br>• below safe level/to unsafe level;<br>• greater risk of disease;<br>• booster dose increases antibody levels/remains above unsafe level;<br>• person immune/can't catch disease;<br>• increased number of memory cells; | Allow faster response | 5 |
| | (e) | Any two from<br>• antibody level higher;<br>• for longer period/decreases more slowly/remain for many years;<br>• levels of antibody increase faster after booster; | | 2 |

Total 15 marks

## Type MS · Classify immunity as natural or artificial and active or passive

### Answer 5 of 7 · Question 3 part b.ii · 2 marks

2023 NOV · Paper 02 · extract 2 of 2 · find part b.ii in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|
| | | | |

Type MS · Classify immunity as natural or artificial and active or passive

Answer 6 of 7 · Question 3 part c · 2 marks

2023 NOV · Paper 02 · extract 1 of 2 · find part c in this mark scheme extract

| Number | | | | |
|---|---|---|---|---|
| 3 | (a) | pathogens; | | 1 |
| | (b) (i) | 1.2 years; | | 1 |
| | (ii) | • active;<br>• artificial; | | 2 |
| | (iii) | Any two from<br>• weakened/dead/attenuated organisms;<br>• toxins;<br>• antigens;<br>• RNA; | Allow named bacterium/virus<br>Ignore genetic material | 2 |
| | (c) | Any two from<br>• passed from mother;<br>• during breastfeeding/from breast milk;<br>• across placenta; | Ignore umbilical cord | 2 |
| | (d) | Any five from<br>• antibody level falls;<br>• below safe level/to unsafe level;<br>• greater risk of disease;<br>• booster dose increases antibody levels/remains above unsafe level;<br>• person immune/can't catch disease;<br>• increased number of memory cells; | Allow faster response | 5 |
| | (e) | Any two from<br>• antibody level higher;<br>• for longer period/decreases more slowly/remain for many years;<br>• levels of antibody increase faster after booster; | | 2 |

Total 15 marks

## Type MS · Classify immunity as natural or artificial and active or passive

### Answer 6 of 7 · Question 3 part c · 2 marks

2023 NOV · Paper 02 · extract 2 of 2 · find part c in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|
| | | | |

Type MS · Classify immunity as natural or artificial and active or passive

Answer 7 of 7 · Question 8 part a.vi · 2 marks

2025 JUN · Paper 02 (regional R) · find part a.vi in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 8 (a) | (i) | cilia/skin/mucus | allow other suitable answers | 1 |
| | (ii) | kills/inhibits growth of pathogens/damage cell membranes; | | 1 |
| | (iii) | • (phagocytes) engulf pathogens/microorganisms / bacteria;<br>• digest/breakdown pathogens/microorganisms / bacteria; | | 2 |
| | (iv) | antigen/immunogen; | | 1 |
| | (v) | • antibodies have a particular shape /are complementary (to antigen);<br>• only recognise/attach/bind to, one particular/specific pathogen / antigen/receptor; | | 2 |
| | (vi) | • natural passive immunity the body receives antibodies naturally;<br>• artificial passive immunity are when antibodies are injected into the body; | accept through breast milk/placenta<br>accept e.g. to give immediate protection against a disease; | 2 |
| (b) | (i) | 360 ÷ 30 = 12;<br>108 ÷ 12;<br>9; | | 3 |
| | (ii) | reduce the risk of disease transmission/infection; | | 1 |
| | (iii) | any two from<br>• improper/overuse of antibiotics/over prescribed;<br>• bacterial DNA mutates/bacteria adapt;<br>• some survive and reproduce | | 2 |
| | | | **Total for Question 8 = 15 marks** | |